

HARSOVA, Romania

WMO: 154060

Lat: 44.6917N

Lon: 27.9636E

Elev: 127

StdP: 14.63

Time Zone: 2.00 (EUE)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	16.1	6.9	7.9	13.8	11.6	10.0	18.6	39.3	17.5	34.7	25.7	7.8	0	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	92.4	69.5	89.9	69.0	87.4	68.5	73.8	85.4	72.0	83.7	70.7	82.2	7.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.7	113.5	77.6	68.8	106.4	75.8	67.4	101.1	74.4	37.4	84.9	35.9	83.1	34.6	82.6	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	21.5	17.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	30.1	36.5	43.3	52.2	63.3	71.5	74.1	73.9	65.3	53.6	44.4	35.1
	DBStd	16.81	8.25	9.02	8.46	7.42	5.65	5.06	4.55	5.16	7.05	7.54	8.82	7.83
	HDD50	2028	615	383	239	63	1	0	0	0	0	51	211	465
	HDD65	5035	1080	798	674	387	97	8	0	3	84	359	618	927
	CDD50	3377	0	6	30	128	414	645	747	740	459	162	43	3
	CDD65	909	0	0	0	3	44	203	283	278	93	5	0	0
	CDH74	8742	0	0	5	60	455	1837	2722	2778	843	41	1	0
	CDH80	3391	0	0	0	7	91	649	1116	1235	290	3	0	0
Wind	WSAvg	6.6	7.3	7.8	8.0	7.6	6.2	5.8	5.5	5.8	5.8	6.1	6.4	7.1
Precipitation	PrecAvg	18.5	1.2	0.8	1.2	1.3	1.9	2.3	2.1	1.5	1.9	1.4	1.3	1.5
	PrecMax	26.3	3.5	2.7	3.0	3.1	5.5	4.1	6.3	4.6	5.0	2.6	3.1	4.0
	PrecMin	10.4	0.1	0.1	0.0	0.1	0.4	0.6	0.4	0.2	0.2	0.1	0.0	0.2
	PrecStd	4.5	0.9	0.6	0.7	0.7	1.3	1.0	1.5	1.1	1.3	0.7	0.8	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.5	64.4	72.1	80.1	86.5	93.4	95.8	96.0	91.8	79.1	70.9	57.3
		MCWB	47.3	53.8	54.2	63.0	66.1	72.7	73.4	69.8	66.1	63.9	60.1	51.6
	2%	DB	47.6	57.2	66.0	73.9	82.1	89.7	91.8	92.5	86.5	74.3	64.6	53.0
		MCWB	44.5	50.6	52.1	58.8	65.2	70.5	70.0	68.9	65.0	60.5	56.9	49.0
	5%	DB	43.8	52.4	61.0	70.1	78.8	86.8	89.3	89.9	82.5	69.9	59.8	48.9
		MCWB	41.9	47.2	49.8	57.1	63.5	69.5	69.4	68.5	64.3	59.0	54.7	46.3
	10%	DB	40.8	48.0	55.9	66.0	75.7	83.6	86.4	87.1	78.2	65.6	56.4	44.8
		MCWB	39.3	44.7	47.5	54.5	62.2	68.3	68.8	67.9	63.4	57.0	52.8	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	55.2	56.1	64.3	69.2	75.4	78.1	75.7	70.9	66.0	61.1	53.5
		MCDB	51.3	63.2	69.1	78.2	81.3	89.0	86.0	87.6	81.2	76.1	68.3	57.1
	2%	WB	44.8	51.0	52.8	61.1	67.4	72.9	73.8	73.2	68.9	62.4	57.9	49.8
		MCDB	47.0	56.3	62.6	72.2	79.1	85.7	84.6	84.9	77.7	69.9	63.0	52.2
	5%	WB	42.4	47.5	50.5	58.4	65.4	71.2	72.2	71.3	67.3	60.6	55.7	46.5
		MCDB	44.2	51.5	58.9	67.8	75.3	83.3	83.0	83.5	76.5	68.3	60.1	49.5
	10%	WB	39.5	44.8	48.1	56.0	63.7	69.7	70.7	70.0	65.3	58.2	53.2	43.0
		MCDB	41.0	47.8	54.2	64.2	72.8	81.4	81.3	82.3	76.9	64.4	56.6	44.8
Mean Daily Temperature Range	5% DB	MDBR	10.9	13.4	16.7	19.4	20.5	21.3	21.5	22.5	20.5	18.0	12.9	10.9
		MCDBR	14.4	20.2	25.5	26.3	25.5	25.4	25.4	27.0	26.1	23.9	18.2	15.8
		MCWBR	11.3	13.0	13.4	13.5	10.9	9.7	8.4	8.8	9.1	11.4	10.9	11.4
	5% WB	MCDBR	14.4	18.9	22.1	24.8	22.7	23.3	22.3	23.6	23.0	21.3	16.6	15.5
		MCWBR	11.5	13.0	12.8	13.3	10.9	9.6	9.2	9.0	9.5	11.3	9.9	11.4
Clear-Sky Solar Irradiance	taub	0.344	0.374	0.407	0.452	0.428	0.430	0.445	0.456	0.425	0.390	0.366	0.345	
	taud	2.293	2.224	2.179	2.067	2.221	2.249	2.217	2.174	2.229	2.307	2.322	2.317	
	Ebn at Noon	238	251	259	257	268	267	261	253	251	243	227	224	
	Edh at Noon	27	34	41	49	44	43	43	44	38	31	26	24	
All-Sky Solar Radiation	RadAvg	414	672	1054	1499	1893	2044	2056	1841	1337	847	467	354	
	RadStd	61	94	106	150	125	127	116	103	100	87	57	48	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	-220	+160	+103	

Nomenclature: See separate page